University of Minnesota Micro Notes

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Contents

1 Introduction

This document serves as a collection of concepts covered in graduate level microeconomics. Specifically, the PhD Microeconomics sequence ECON 8101 (Werner), ECON 8102 (Phelan), ECON 8103 (Rustichini), and ECON 8104 (Rahman) at the University of Minnesota. If you ever encouter these notes and have questions or spot an error, be sure to reach out to me here. Note that everything below is very personalized. I will officially state some theorems but the main goal of writing everything out is to cement my own intuition. Most everything will include statements to help intimidating things feel a bit more friendly.

2 ECON 8101 (Werner)

The main units/concepts that have been covered thus far have been Production Sets, Production Functions, Profit Maximization, Convex Analysis and Duality, Supply and Profit, Preferences and Utility Functions, Walrasian Demand, the Slutsky Matrix, Revealed Preference, Theorem of Topkis. I will discuss the main ideas and theorems that pervade the aforementioned concepts.

2.1 Production

Definition 1 (Supermodularity). *A function f is supermodular if for any x, y we have that*

$$f(x \vee y) - f(x) \geq f(x \wedge y) - f(y).$$

This also implies the complementary of inputs (which intuitively is something that is indeed defined by the production function and not an inherent chracteristic of the inputs themselves).

There are two more key concepts in this section: Profit Maximization and Cost Minimization. The concepts are simple enough and likely won't come up much on the midterm so I'm just going to note down the two relevant definitions.

Definition 2 (Profit Function). *Given a price vector $p \in \mathbb{R}^n$, the profit function π^* is*

$$\pi^*(p) = \sup_{y \in Y} py.$$

Note here that Y is a production set which means it is usually structured as a vector of inputs and one ouput stacked on top of eachother where the inputs are usually normalized to be negative.

Definition 3 (Cost Function). *Given a vector of prices $p \in \mathbb{R}^n_+$, output level $z \in \mathbb{R}$, and production function f, the cost function C^* is*

$$\begin{aligned} C^*(w, z) = \min_{x \geq 0} \quad & px \\ s.t. \quad & f(x) \geq z \end{aligned}$$

The conditional factor demand correspondence x^* is defined as $C^*(p,z) \equiv px^*(p,z)$ or can also be seen as

$$x^*(w,z) = \arg\min_{x \geq 0} \quad px$$
$$s.t. \quad f(x) \geq z$$

I need to be comfortable with working with these optimization objects. They are not things that you can do simple algebra with but require more nuanced arguments that take advantage of inequalities and properties of the maxima and minima. There are often many problems regading these concepts, however, that try to make one reason about

2.2 Consumer Theory

Definition 4 (Indirect Utility Function)**.** *The function u^* such that given a price vector $p \in \mathbb{R}^n_+$ and income $w \in \mathbb{R}_{++}$*

$$u^*(p,w) = \max_{x \geq 0} \quad u(x)$$
$$s.t. \quad px \leq w$$

Definition 5 (Walrasian Demand Function)**.** *The function x^* such that given a price vector $p \in \mathbb{R}^n_+$ and income $w \in \mathbb{R}_{++}$*

$$x^*(p,w) = \arg\max_{x \geq 0} \quad u(x)$$
$$s.t. \quad px \leq w$$

Note that $u^*(p,w) = u(x^*(p,w))$.

I want to dedicate a small section to talk about the Walrasian demand function and the indirect utility function. The key intuition to keep in mind is Walras law, that given prices and income, the optimal consumption bundle consumes all of the income which is something that is guaranteed when a utility function is locally non-satiated. This is a necessary condition of a bundle that optimized utility, not sufficient. It is only a relation between optimal bundles and their corresponding prices not that a bundle that satisfies that price relation is optimal. This is something that often trips me up. If we want to make statements about similar consumption bundles by price then we have to use the definition, not the price relation that lns gives us.

Theorem 2.1 (Generalized Weak Axiom of Revealed Preference - GWARP)**.** *Given two price vectors $p^a, p^b \in \mathbb{R}^n_{++}$ and consumption bundles $x^a, x^b \in \mathbb{R}^n_+$. If we have that*

$$p^a x^b \leq p^a x^a$$

then

$$p^b x^b \leq p^b x^a$$

follows. Tangent to this GWARP for Walrasian demand functions. Specifically, given prices $p, p' >> 0$ *and incomes* w, w' *we have that*

$$px^*(p', w') \leq w \implies p'x^*(p, w) \geq w'.$$

Intuitively we can think of this as analyzing the preference behavior of individuals by looking at what types of bundles they pick (hence the "revealed prefence"). For the Walrasian demand function, think of it as; if a given utility maximizing consumption bundle (under prices and income basket a) is affordable given prices and income basket b , then the utility maximizing consumption bundle under prices and income basket b is not affordable given prices and income basket a.

The strong axiom of revealed preference is an argument in transitivity. If we have a chain of consumption bundles whose preferences have been "revealed" then we can relate the first and $n-$th bundle in that chain.

Theorem 2.2 (The Law of Compensated Demand)**.** *Let* x^* *be a Walrasian demand function of a consumer with continuous and locally non-satiated utility function. Then for every* $p >> 0, w > 0, p' >> 0, w' > 0$ *such that* $w' = p'x^*(p, w)$ *we have that*

$$[x^*(p', w') - x^*(p, w)][p' - p] \leq 0.$$

To be completely honest, I'm lacking in my intuition here. I think something like: if the optimal consumption bundle is affordable at a different price set then either the other bundles prices are lower or the other bundle is smaller.

Definition 6 (The Slutsky Matrix)**.** *Given a differentiable Walrasian demand function* x^* *we define the Slutsky matrix by element as*

$$s_{k\ell} = \frac{\partial x_k^*(p, w)}{\partial p_\ell} + \frac{\partial x_k^*(p, w)}{\partial w} x_\ell^*(p, w).$$

Law of compensated demand implies that this matrix is negative semi-definite and the matrix is also symmetric (we are often asked to confirm these facts).

2.3 Comparative Statics

The main concept to keep in mind here is the Theorem of Topkis. The motivation here is that we can use this Theorem to make statements about the behavior of certain functions with respect to certain variables. For example, say we have some input demand function $x^*(p, w, t)$ that is a function of input prices, output prices, and some tax rate and we want to analyze the behavior (non-decreasing, non-increasing) with respect to a to any of the tax rate or inputs, we can apply Topkis. On onset, it personally feels a little specialized, but it has the flavor of something that is subtly very powerful.

Theorem 2.3 (Theorem of Topkis). *Consider a function $f : X \times T \to R$ and the corresponding function*

$$\phi(t) = \arg\max_{x \in X} f(x, t).$$

We can say that ϕ is nondecreasing (nonincreasing) in t on S if

1. *S is a lattice (the $\vee$ and $\wedge$ operations are included in S).*

2. *$f(x, t)$ is supermodular in x for every t*

3. *$f(x, t)$ has nondecreasing (nonincreasing) difference in t. Aka for $t' > t$ and $x' > x$ we have*

$$f(x', t') - f(x', t) \geq (\leq) f(x, t') - f(x, t).$$

2.4 Uncertainty and Utility

Now we start discussing the interplay between preferences / utility functions and risk. Before that, there are a handful of key definitions to consider.

Definition 7 (State Separable Utility Representation). *We say that a preference $\succeq$ has a state-separable utility representation, if there exist utility functions $v_s : \mathbb{R}_+ \to \mathbb{R}$ for all s, such that*

$$c \succeq \iff \sum_{s=1}^{S} v_s(c_s) \geq \sum_{s=1}^{S} v_s(c'_s),$$

for every $c, c' \in \mathbb{R}^S_+$.

Definition 8 (Expected Utility Representation). *We say that $\succeq$ has an expected utility representation with respect to probabilities $\{\pi_s\}$, if there exists a function $v : \mathbb{R}_+ \to \mathbb{R}$ such that*

$$c \succeq c' \iff \sum_{s=1}^{S} \pi_s v(c_s) \geq \sum_{s=1}^{S} \pi_s v(c'_s),$$

for every $c, c' \in \mathbb{R}^S_+$. We call the function v in this case as the <u>von Neumann-Morgenstern (or Bersnoulli) utility</u>

I'm not going to cover proofs here that won't be incredibly focused on during the prelims. One important theorem to consider, however, is the existence of a state separable utility representation.

Definition 9 (Axiom of Irrelevance of Common Consequences). *Take any $c \in \mathbb{R}^S_+$ and $y \in \mathbb{R}$ and write*

$$c_{-s}y = (c_1, c_2, \ldots, y, c_{s+1}, c_{s+2}, \ldots, c_S).$$

That it is, $c_{-s}y$ is the vector c that has replaced c_s with y. A preference relation satisfies the ICC axiom if for all $c, d \in \mathbb{R}^S_+$ and any $y, w \in \mathbb{R}_+$ the following condition holds

$$c_{-s}y \succeq d_{-s}y \Longleftrightarrow c_{-s}w \succeq d_{-s}w.$$

Theorem 2.4 (Theorem 16.1)**.** *If $S \geq 3$ and the preference relation $\succeq$ is strictly increasing and continuous. Then $\succeq$ has a state-separable utility representation iff it satisfies the ICC axiom.*

Now consider probabilities of states $\{\pi_s\}$such that $\pi_s > 0$ for each s. Additionally, we denote $E(c) = \sum_s \pi_s c_s$ as the expected value of $c = (c_1, \ldots c_S)$ and $\mathbf{E}(c) = (E(c), \ldots, E(c))$ as corresponding risk-free consumption plan.

Definition 10 (Risk Aversion)**.** *We say that preference relation $\succeq$ is risk averse with respect to $\{\pi_s\}$, if*

$$\mathbf{E}(c) \succeq c$$

for every c.

Theorem 2.5 (Theorem 16.2)**.** *If $S \geq 3$ and the preference relation $\succeq$ is strictly increasing and continuous. Then $\succeq$ satisfies the ICC Axiom and is risk averse with respect to probabilities $\{\pi_s\}$ if and only if it has a concave expected utility representation with respect to $\{\pi_s\}$.*

Definition 11 (Ellsberg Paradox)**.** *An urn has 90 balls of which 30 are red and the rest are blue and yellow. Exact numbers of blue balls and yellow balls are not known. Consider bets of \$1 on a ball of a certain color (or colors) drawn from the urn. Denote bets by 1_R, 1_B, $1_{R\vee Y}$, etc. Typical preferences over bets are*

$$1_R \succ 1_B, \qquad 1_{B\vee Y} \succ 1_{R\vee Y}$$

This pattern of preferences is incompatible with expected utility: it cannot be that $\pi(R) > \pi(B)$ and $\pi(B \vee Y) > \pi(R \vee Y)$, because $\pi(B \vee Y) = \pi(B) + \pi(Y)$ holds for any probability measure π.

The intuition here is that there are various situations where common characteristics like transitivity and separability for preferences over lotteries fail because humans have preferences that tend to be risk averse. As such, in an attempt to write more consistent utility representations, we employ the use of something like below.

Definition 12 (Multiple-Prior Expected Utility)**.** *An alternative to expected utility and one that can explain the Ellsberg paradox is the* ***multiple-prior expected utility.*** *It takes the form*

$$\min_{P \in \mathcal{P}} E_P[v(c)], \tag{mpeu}$$

where $v : R_+ \to R$ is the von Neumann–Morgenstern utility (with no date-0 consumption) and $\mathcal{P}$ is a convex and closed set of probability measures on $\mathcal{S}$.

The set of probability measures $\mathcal{P}$ reflects the agent's ***ambiguous beliefs****.*

Next, let's try to tie together probabilities and preferences directly. Let $Z = \{z_1, \ldots, z_K\}$ be a finite set of outcomes. A **lottery** is a probability distribution on Z. It is an assignment of probabilities $\{p_i\}_{i=1}^K$ where p_i is the probability of winning outcome z_i. Agents can have pereferences over lotteries and

Definition 13 (Expected Utility Representation)**.** *Preference relation $\succeq$ on the set of lotteries $\mathcal{L}$ has an expected utility representation if there exists a function $v : Z \to \mathbb{R}$ such that*

$$L \succeq L' \iff \sum_{i=1}^{K} p_i v(z_i) \geq \sum_{i=1}^{K} p_i' v(z_i).$$

Definition 14 (Definition 18.1)**.** *$\tilde{z}$ first-order stochastically dominates $\tilde{y}$ if*

$$F_z(t) \leq F_y(t), \qquad \forall t \in [a, b]. \tag{1}$$

Theorem 2.6 (Theorem 18.2)**.** *$\tilde{z}$ first-order stochastically dominates $\tilde{y}$ if and only if*

$$E[v(\tilde{z})] \geq E[v(\tilde{y})]$$

for every nondecreasing continuous v.

That is, $\tilde{z}$ FSD $\tilde{y}$ if and only if every expected-utility maximizing agent with nondecreasing utility prefers $\tilde{z}$ to $\tilde{y}$.

Definition 15 (Definition 18.4)**.** *$\tilde{z}$ second-order stochastically dominates $\tilde{y}$ if*

$$\int_a^w F_z(t)\, dt \leq \int_a^w F_y(t)\, dt, \qquad \forall w \in [a, b]. \tag{2}$$

If $\tilde{z}$ SSD $\tilde{y}$, then $E(\tilde{z}) \geq E(\tilde{y})$. Also, if $\tilde{z}$ FSD $\tilde{y}$, then $\tilde{z}$ SSD $\tilde{y}$.

Theorem 2.7 (Theorem 18.5)**.** *$\tilde{z}$ second-order stochastically dominates $\tilde{y}$ if and only if*

$$E[v(\tilde{z})] \geq E[v(\tilde{y})]$$

for every nondecreasing concave continuous v.

That is, $\tilde{z}$ SSD $\tilde{y}$ if and only if every agent with risk averse nondecreasing expected utility prefers $\tilde{z}$ to $\tilde{y}$. If $\tilde{z}$ SSD $\tilde{y}$ and $\tilde{z}$ and $\tilde{y}$ have the same expectation $E(\tilde{z}) = E(\tilde{y})$, then we say that $\tilde{y}$ is **more risky** than $\tilde{z}$.

2.5 Recursive Utility

This section usually corresponds to one specific type of prelim problem (2.7.1): we are asked to define families of recursive utility functions, define dynamic consistency and then show a handful of properties of a family of recursive utility functions that can be represented with some aggregator funciton.

To start at the beginning, however, consider one of the most common representation of utility. The discounted utility function

$$\sum_{t=0}^{\infty} \beta^t u(c_t).$$

From here, we can define

Definition 16 (Continuation Utility Function)**.** *A function which assigns utility for every consumption plan starting at date-t, denoted by $c^t = (c_t, c_{t+1}, \ldots)$, for every $t \geq 1$. Specifically, it is*

$$U_t(c^t) = \sum_{\tau=t}^{\infty} \beta^{\tau-t} u(c_\tau).$$

There are several properties of continuation functions and collections of continuation functions that are worth discussing.

Definition 17 (Dynamic Consistency)**.** *A family of continuation functions is dynamically consistent if for every t, we have that for every $c, d \in \mathbb{R}_+^\infty$ such that $c_\tau = d_\tau$ for $\tau < t$*

$$U_0(c) \geq U_0(d) \iff U_t(c^t) \geq U_t(d^t).$$

Definition 18 (Recursive Utility and Stationarity)**.** *More generally, a family of utility functions $\{U_t\}$ is recursive if*

$$U_t(c^t) = G(u(c_t), U_{t+1}(c^{t+1}))$$

for every c and t and some aggregator function $G : \mathbb{R}^2 \to \mathbb{R}$ and in-period utility function $u : \mathbb{R}_+ \to \mathbb{R}$

A recursive family of utility functions is stationary if $U_t = U$ for every t. That is

$$U(c^t) = G(u(c_t), U(c^{t+1})).$$

Definition 19 (Preference for Early Consumption)**.** *A recursive family of utility functions $\{U_t\}$ exhibits preference for early consumption if for every $c \in C$, $w, v \in \mathbb{R}_+$ such that $w > v$, and every t, it is the case that*

$$U_t((c_1, \ldots, c_{t-1}, w, v, c_{t+2}, \ldots)) > U_t((c_1, \ldots, c_{t-1}, v, w, c_{t+2}, \ldots)).$$

Werner also talks a little about recursive utility under uncertainty, but in my experience, this has never showed up on either a HW, Midterm, Final, or Prelim problem. The concept is generally, not all that different.

Now we must also define a finite set of possible sets at any given time S. The product set S^∞ represents the set of all possible sequences of these states over infinite time. Also consider some probability measure Π over S^∞.

Definition 20 (Discounted Expected Utility)**.** *As discussed above, it is a function which assigns utility for every consumption plan starting at date-t, denoted by $c^t = (c_t, c_{t+1}, \ldots)$, for every $t \geq 1$. Specifically, it is*

$$U_t(c^t) = \sum_{\tau=t}^{\infty} \beta^{\tau-t} \mathbb{E}_t[u(c_\tau)].$$

Definition 21 (State Dependent Recursive Utility and Stationarity)**.** *A state dependent utility function $U : S \times C \to \mathbb{R}$ is stationary recursive if it satisfies*

$$U(s_t, c^t(s^t)) = F(c_t(s^t), \mu_{s_t}(U(c^{t+1}))), \quad \forall s_t, t$$

for every $c \in C$, where $F : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}$ is the aggregator and $\mu_s : \mathbb{R}^S \to \mathbb{R}$ is the state dependent certianty equivalent.

Although it is quite unlikely that there will be a question related to Epstein-Zin-Weil preferences. This is quite relevant for a lot of work that is being done in macro.

Definition 22 (Epstein-Zin-Weil)**.** *A recursive utility satisfying*

$$U_t(c^t) = \left(c_t^\alpha + \beta[\mathbb{E}_t(U_{t+1}(c^{t+1}))^{1-\rho}]^{\alpha/(1-\rho)}\right)^{1/\alpha}$$

for $\alpha =\neq 0$ and $\rho \neq 1$.

2.6 Social Choice

This section is very tame for Werner relative to everything else. There are rarely any questions regarding it. Consider a world with

- X, a set of social alternatives
- N, a set of individuals ($N = \{1, 2, \ldots, n\}$).
- $\succeq_i \in \mathcal{R}$, a preference relation of individual i on X and $\mathcal{R}$ is the set of all such preference relations. Let $\mathcal{P}$ denote the subset of $\mathcal{R}$ such that there are no indifferences.

With these objects in hand, we can then define

Definition 23 (Social Welfare Function)**.** *A function $F\mathcal{P}^n \to \mathcal{R}$ that assigns a (social) preference relationto every profile of individual preferences. Specifically, we denote this as*

$$\succeq_F \equiv F(\succ_1, \ldots, \succ_n).$$

Arrow's theorem is one of the groundbreaking results in economics that everyone is familiar with. We will look at the proof more in depth in the Rahman section, but there is a quick statement of it. First consider the following two qualities that a social welfare function F may have

- (Pareto) For every $x, y \in X$ and every profile $(\succ_1, \ldots, \succ_n)$, if $x \succ_i y$ for all i, then $x \succ_F y$.
- (Independence of Irrelevant Alternatives, IIA) For every $x, y \in X$ and every pair of profiles $(\succ_1, \ldots, \succ_n)$ and $(\succ_1', \ldots, \succ_n')$, it holds

$$\text{if } x \succ_i y \text{ iff } x \succ_i' y \text{ for every } i, \text{ then } x \succeq_F \text{ iff } x \succeq_F' y.$$

Theorem 2.8 (Arrow's Theorem). *If there are at least three alternative (i.e. $|X| \geq 3$) and the social welfare function F is both Pareto and satisfies IIA, then F is dictatorial.*

That is, there exists some i^* such that if $x \succ_{i^*} y$ then $x \succ_F y$ for every $x, y \in X$ and every profile $(\succ_1, \ldots, \succ_n)$.

2.7 Exercises

I didn't cover optimal portfolio questions in the content above but refer to problems (2.7.2) for content on them.

2.7.1 Prelim: Spring 2025

Consider a family of utility functions $\{U_t\}_{t=0}^{\infty}$ defined on the set ℓ_+^∞ of bounded, positive consumption plans of a single good over infinite time horizon without uncertainty.

(a) Define what it means for the utility functions $\{U_t\}$ to be recursive. Define what it means for $\{U_t\}$ to be dynamically consistent.

SOLUTION: $\{U_t\}$ is **recursive** if we can write

$$U_t(c_t) = G(u(c_t), U_{t+1}(c^{t+1})),$$

for every c and t and some aggregator function $G : \mathbb{R}^2 \to \mathbb{R}$ and period-utility function $u : \mathbb{R}_+ \to \mathbb{R}$.

It is **dynamically consistent** if for every t, and $c, d \in \mathbb{R}_+^\infty$ such that $c_\tau = d_\tau$ for any $\tau < t$,

$$U_0(c) \geq U_0(d) \iff U_t(c^t) \geq U_t(d^t).$$

■

(b) Consider a stationary recursive utility function with aggregator

$$G(z_1, z_2) = z_1 + \beta \ln(z_2 + 1),$$

for $z_1 \geq 0$ and $z_2 \geq 0$, where $0 < \beta < 1$, and linear period-utility $u(x) = x$. Show that this recursive utility function is dynamically consistent.

SOLUTION: Pick any arbitrary $t \geq 0$. Using this, select any arbitrary $c, d \in \mathbb{R}_+^\infty$ such that $c_\tau = d_\tau$ for any $\tau < t$. We can write for any $\tau < t$

$$\begin{aligned}
& U_\tau(c^\tau) \geq U_\tau(d^\tau) \\
\iff & c_\tau + \beta \ln\left(U_{\tau+1}(c^{\tau+1}) + 1\right) \geq d_\tau + \beta \ln\left(U_{\tau+1}(d^{\tau+1}) + 1\right), \\
\iff & \beta \ln\left(U_{\tau+1}(c^{\tau+1}) + 1\right) \geq \beta \ln\left(U_{\tau+1}(d^{\tau+1}) + 1\right), \\
\iff & \ln\left(U_{\tau+1}(c^{\tau+1}) + 1\right) \geq \ln\left(U_{\tau+1}(d^{\tau+1}) + 1\right), \\
\iff & U_{\tau+1}(c^{\tau+1}) + 1 \geq U_{\tau+1}(d^{\tau+1}) + 1, \\
\iff & U_{\tau+1}(c^{\tau+1}) \geq U_{\tau+1}(d^{\tau+1}).
\end{aligned}$$

Since we used double implication statements the whole way, we can inductively use this fact to generally state that

$$U_s(c^s) \geq U_s(d^s) \iff U_k(c^k) \geq U_k(d^k),$$

for any $s, k \leq t$. Pick $s = 0$ and $k = t$ and this is exactly the definition of dynamic consistency for our utility function. ■

(c) A stationary recursive utility function U is said to exhibit preference for early consumption if, for any consumption plan that yields w at date t and y at date $t+1$, with $w > y$, the agent prefers it to an otherwise identical plan (i.e., consumption is the same at all other dates) that delivers y at date t and w at date $t+1$. State this definition formally. Show that the recursive utility function of part (b) exhibits preference for early consumption.

SOLUTION: A recursive family of functions exhibits a preference for early consumption if for every $c \in C$, $w, y \in \mathbb{R}_+$ with $w > y$, and $t \geq 0$ we have that

$$U_t(\hat{c}^t) > U_t(\bar{c}^t)$$

where $\hat{c} = (c_1, \ldots, c_{t-1}, w, y, c_{t+2}, \ldots)$ and $\bar{c} = (c_1, \ldots, c_{t-1}, y, w, c_{t+2}, \ldots)$.

To prove this for the recursive utility function from before, pick any $c \in C$, $w, y \in \mathbb{R}_+$ with $w > y$, and $t \geq 0$. We can then write out

$$\begin{aligned}
& U_t(\hat{c}^t) > U_t(\bar{c}^t), \\
\iff & w + \beta \ln(U_{t+1}(\hat{c}^{t+1}) + 1) > y + \beta \ln(U_{t+1}(\bar{c}^{t+1}) + 1), \\
\iff & w + \beta \ln(\beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + y + 1) > y + \beta \ln(\beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w + 1).
\end{aligned}$$

From here we continue

$$
\begin{aligned}
&\iff w - y > \beta \ln(\beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w + 1) - \beta \ln(\beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + y + 1), \\
&\quad \iff 1 > \frac{\beta \left[\ln(\beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w + 1) - \ln(\beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + y + 1)\right]}{w - y}, \\
&\quad \iff 1 > \frac{\beta \left[\ln(w + A) - \ln(y + A)\right]}{w - y},
\end{aligned}
$$

where $A = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + 1 = \beta \ln(U_{t+2}(\hat{c}^{t+2}) + 1) + 1$. From here, using mean value theorem, we can write

$$
\frac{\beta \left[\ln(w + A) - \ln(y + A)\right]}{w - y} < \frac{\beta}{w + A} \leq \beta < 1.
$$

Since we used double implications the whole way, we have shown that

$$
U_t(\hat{c}^t) > U_t(\bar{c}^t)
$$

and indeed this family of utility functions exhibits preference for early consumption.

One note is that instead of using the mean value theorem, we can use the property that $\ln(1 + x) < x$ for all $x > 0$. Specifically, we can write $B = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1)$, which we know is greater than 0, and then define $C_w = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + w$ and $C_y = \beta \ln(U_{t+2}(\bar{c}^{t+2}) + 1) + y$. This let's us write both of the expressions

$$
\begin{aligned}
\ln(1 + C_w) < C_w &\implies \beta \ln(1 + C_w) < C_w \\
\ln(1 + C_y) < C_y &\implies \beta \ln(1 + C_y) < C_y
\end{aligned}
$$

Subtracting these two expressions gives

$$
w - y > \beta[\ln(1 + C_w) - \ln(1 + C_y)],
$$

which is exactly the expression we needed to prove above. ■

(d) Derive the marginal rate of intertemporal substitution between consumption at dates $t + 1$ and t, and demonstrate that it is strictly less than 1 for every consumption plan.

Solution: Since this utility function is a recursive one and also linearly separable, we can pick any consumption plan $c \in C$ and write

$$
MRS_{t+1,t} = \frac{\partial U_t(c^t)/\partial c_{t+1}}{\partial U_t(c^t)/\partial c_t},
$$

$$
\begin{aligned}
&= \frac{\frac{1}{\partial c_{t+1}}\left[G(c_t, G(c_{t+1}, U_{t+2}(c^{t+2}))\right]}{\frac{1}{\partial c_t}\left[G(c_t, G(c_{t+1}, U_{t+2}(c^{t+2}))\right]},\\
&= \frac{\frac{1}{\partial c_{t+1}}\left[c_t + \beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right]}{\frac{1}{\partial c_t}\left[c_t + \beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right]},\\
&= \frac{1}{\partial c_{t+1}}\left[c_t + \beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right],\\
&= \frac{1}{\partial c_{t+1}}\left[\beta \ln(1 + G(c_{t+1}, U_{t+2}(c^{t+2})))\right],\\
&= \frac{1}{\partial c_{t+1}}\left[\beta \ln(1 + c_{t+1} + \beta \ln(U_{t+2}(c^{t+2}) + 1))\right],\\
&= \frac{\beta}{1 + c_{t+1} + \beta \ln(U_{t+2}(c^{t+2}) + 1)},\\
&\leq \frac{\beta}{1 + c_{t+1}},\\
&\leq \beta,\\
&< 1.
\end{aligned}
$$

■

2.7.2 Prelim: Fall 2024

There are two assets: a risk-free asset with return r_f and a risky asset with return $\tilde{r}$. Returns are per-dollar returns, meaning they represent the total return for one dollar invested. The agent has an expected utility function with von Neumann (or Bernoulli) utility index

$$v(x) = -(\alpha - x)^2,$$

for $\alpha > 0$, and initial wealth $w > 0$. Assume that $\alpha > wr_f$. Negative investment (i.e., short selling) is allowed for both assets.

(a) Find the optimal investment in the risky asset as a function of the expected return and the variance of the return on the risky asset, the risk-free return, and the agent's wealth.

SOLUTION: Given wealth $w > 0$, the agent experiences the following maximization problem

$$\max_{w_r} \quad \mathbb{E}\left[v(r_f(w - w_r) + \tilde{r}w_r)\right].$$

Here is the corresponding first order condition

$$\mathbb{E}\left[\tilde{r}v'(r_f(w - w_r) + \tilde{r}w_r)\right] = 0,$$

$$
\begin{aligned}
\mathbb{E}\left[\tilde{r}(-2)(\alpha - r_f(w - w_r) - \tilde{r}w_r)\right] &= 0,\\
\mathbb{E}\left[\tilde{r}\alpha - \tilde{r}r_f(w - w_r) - \tilde{r}^2 w_r)\right] &= 0,\\
\mathbb{E}[\tilde{r}\alpha] - \mathbb{E}[\tilde{r}r_f(w - w_r)] - \mathbb{E}[\tilde{r}^2 w_r] &= 0,\\
\alpha\mathbb{E}[\tilde{r}] - r_f(w - w_r)\mathbb{E}[\tilde{r}] - w_r\mathbb{E}[\tilde{r}^2] &= 0,\\
w_r(\mathbb{E}[\tilde{r}] - \mathbb{E}[\tilde{r}^2]) &= \mathbb{E}[\tilde{r}](wr_f - \alpha),\\
w_r &= \frac{\mathbb{E}[\tilde{r}](wr_f - \alpha)}{\mathbb{E}[\tilde{r}] - \mathbb{E}[\tilde{r}^2]}.
\end{aligned}
$$

Finally, we can write

$$
\boxed{w_r = \frac{\mathbb{E}[\tilde{r}](\alpha - wr_f)}{Var(\tilde{r}) + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}]}.}
$$

■

Consider the following comparative statics exercise. The return $\tilde{r}$ is changed to a more risky return $\tilde{r}'$. The expected return of $\tilde{r}'$ is the same as of $\tilde{r}$, that is, $E(\tilde{r}') = E(\tilde{r})$.

(b) Prove that $\mathrm{var}(\tilde{r}') \geq \mathrm{var}(\tilde{r})$, where var denotes the variance.

Solution: Formally, from the definition of a riskier asset, we know that

$$
\mathbb{E}[v(\tilde{r}')] \geq \mathbb{E}[v(\tilde{r})].
$$

Let's go from here then

$$
\begin{aligned}
\mathbb{E}[v(\tilde{r}')] &\leq \mathbb{E}[v(\tilde{r})],\\
\mathbb{E}[-(\alpha - \tilde{r}')^2] &\leq \mathbb{E}[-(\alpha - \tilde{r})^2],\\
\mathbb{E}[(\alpha - \tilde{r}')^2] &\geq \mathbb{E}[(\alpha - \tilde{r})^2],\\
\mathbb{E}[(\tilde{r}' - \mathbb{E}[\tilde{r}])^2] &\geq \mathbb{E}[(\tilde{r} - \mathbb{E}[\tilde{r}])^2],\\
\mathbb{E}[(\tilde{r}' - \mathbb{E}[\tilde{r}'])^2] &\geq \mathbb{E}[(\tilde{r} - \mathbb{E}[\tilde{r}])^2],\\
Var(\tilde{r}') &\geq Var(\tilde{r}).
\end{aligned}
$$

■

(c) Suppose that $E(\tilde{r}) > r_f$. Show that the optimal investment in the risky asset with more risky return $\tilde{r}'$ is smaller than the optimal investment with less risky return $\tilde{r}$, all else unchanged.

Solution: We know from part (a) that the optimal investments in the less risky and more risky asset are respectively

$$w_r = \frac{\mathbb{E}[\tilde{r}](\alpha - wr_f)}{Var(\tilde{r}) + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}]}, \quad w_{r'} = \frac{\mathbb{E}[\tilde{r}'](\alpha - wr_f)}{Var(\tilde{r}') + (\mathbb{E}[\tilde{r}'])^2 - \mathbb{E}[\tilde{r}']}.$$

To make the comparison, consider the sequence of double implications

$$\begin{aligned}
w_r &\geq w_{r'}, \\
\frac{\mathbb{E}[\tilde{r}](\alpha - wr_f)}{Var(\tilde{r}) + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}]} &\geq \frac{\mathbb{E}[\tilde{r}'](\alpha - wr_f)}{Var(\tilde{r}') + (\mathbb{E}[\tilde{r}'])^2 - \mathbb{E}[\tilde{r}']}, \\
\frac{\mathbb{E}[\tilde{r}](\alpha - wr_f)}{Var(\tilde{r}) + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}]} &\geq \frac{\mathbb{E}[\tilde{r}](\alpha - wr_f)}{Var(\tilde{r}') + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}]}, \\
\frac{1}{Var(\tilde{r}) + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}]} &\geq \frac{1}{Var(\tilde{r}') + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}]}, \\
Var(\tilde{r}) + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}] &\leq Var(\tilde{r}') + (\mathbb{E}[\tilde{r}])^2 - \mathbb{E}[\tilde{r}], \\
Var(\tilde{r}) &\leq Var(\tilde{r}').
\end{aligned}$$

Since the last line is true (as we showed in part (c)), it must be the case that $w_{r'} < w_{r'}$. ■

Suppose that the return $\tilde{r}$ second-order stochastically dominates $\tilde{r}'$ rather than being more risky. In this case, do not assume that the expected returns are equal.

(d) Does the argument from part (c), which shows that the optimal investment with return $\tilde{r}'$ is smaller than the optimal investment with $\tilde{r}$, still hold? Explain why or why not.

Solution: In this case the argument does not hold. It could be the case that the $\tilde{r}$ second-order stochastically dominates $\tilde{r}'$ but the expected return $\mathbb{E}[\tilde{r}']$ is so much higher than the expected return from the risk-less asset $\mathbb{E}[\tilde{r}]$ that it overcomes the agent's aversion to risk. ■

3 ECON 8102 (Phelan)

Phelan's class is very good for grounding intuition about competitive equilibrium. We don't cover as many topics as in the later mini's, which gives one a lot of time to really understand what's going on. Additionally, he doesn't publish any official set of notes, so there are likely going to be some holes.

3.1 Economy Primitives

Before we dive into the meat of the content of this course, we need to start with some basic primitives. I'll set up two types of economies as contextual environments and then dive into specifics about everything.

Pure Endowment Economies: Economy in which there exists only households who are bestowed an endowment of resources by some unseen power. There are

- A set of agents, $\mathcal{I} = \{1, \ldots, I\}$ for some $I \in \mathbb{N}$.
- A set of goods/commodities, $\mathcal{M} = \{1, \ldots, M\}$ for some $M \in \mathbb{M}$.
- Each agent i has a consumption set $C_i \subseteq \mathbb{R}^m$ which indicates the possible set of bundles that agent i can consume.
- Endowments of each good for each agent, $e = \{\{e_{i,m}\}_{m \in \mathcal{M}}\}_{i \in \mathcal{I}}$, $e_{i,m} \geq 0$.
- Allocation: consumption allocations for agents, $c = \{\{c_{i,m}\}_{m \in \mathcal{M}}\}_{i \in \mathcal{I}}$, $c_{i,m} \in \mathbb{R}$.

Production Economies: Economy similar to above but there are now also firms who are able to produce consumption goods as well.

- A set of agents, $\mathcal{I} = \{1, \ldots, I\}$ for some $I \in \mathbb{N}$.
- A set of firms, $\mathcal{H} = \{1, \ldots, H\}$ for some $H \in \mathbb{N}$.
- A set of goods/commodities, $\mathcal{M} = \{1, \ldots, M\}$ for some $M \in \mathbb{M}$.
- Each agent i has a consumption set $C_i \subseteq \mathbb{R}^m$ which indicates the possible set of bundles that agent i can consume.
- Each firm i has a production set $Y_h \subseteq \mathbb{R}^m$ which indicates the possible set of bundles that firm h can produce.
- Ownerships of person i in firm h: $\theta_{i,h} \in [0, 1]$ for all i, h.
- Endowments of each good for each agent, $e = \{\{e_{i,m}\}_{m \in \mathcal{M}}\}_{i \in \mathcal{I}}$, $e_{i,m} \geq 0$.
- Allocation: tuple of consumption allocations and firm production plans,

  $$(c, y) = (\{\{c_{i,m}\}_{m \in \mathcal{M}}\}_{i \in \mathcal{I}}, \{\{y_{h,m}\}_{m \in \mathcal{M}}\}_{h \in \mathcal{H}})$$

  for $c_{i,m} \in \mathbb{R}$, $y_{h,m} \in \mathbb{R}$.

With the setups above in mind, here are a couple of definitions that will likely come up in problems.

Definition 24 (Utility Possibilities Set)**.** *The set of attainable vectors of utility levels. That is*

$$U = \left\{ \{u_i(c_i)\}_{i \in \mathcal{I}} :, c_i \in C_i, \sum_i c_{i,m} \leq \sum_i e_{i,m}, \forall m \right\}.$$

Definition 25 (Utility Possibilities Frontier)**.** *Let U be the utility possibilities set. The UPF is the set*

$$\left\{ \{u_i(c_i)\}_{i \in \mathcal{I}} \in U : \nexists (u_1', u_2') \in U \quad s.t. \quad u_i' > u_i, \forall i \right\}.$$

Definition 26 (Better than Sets)**.** *The Better than Set, $B(c)$, for a given consumption bundle c is the set of all consumption bundles that an agent prefers. [I NEED TO CHECK OVER THIS BECAUSE PHELAN ASKS FOR SOMETHING DIFFERENT IN THE HWS]*

Finally, we say that a specific allocation is **feasible** if the total amount of a specific good consumed across all agents is less than the sum of the endowment of that good across every agent (and production of that good across all firms). To put this in more specific terms, for endowment economies, an allocation c, given endowments e is feasible if $c_i \in C_i$ for all $i \in \mathcal{I}$ and for all $m \in \mathcal{M}$

$$\sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_i e_{i,m}.$$

For production economies, an allocation (c, y), given endowments e is feasible if if $c_i \in C_i$ for all $i \in \mathcal{I}$, $y_h \in Y_h$ for all $h \in \mathcal{H}$, and for all $m \in \mathcal{M}$

$$\sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_i e_{i,m} + \sum_{h \in \mathcal{H}} y_{h,m}.$$

3.2 Efficiency

You should not be surprised that we are once again talking about Pareto efficiency. It is, however, very straightforward in the Phelan barebones world.

Definition 27 (Pareto Dominating Allocation)**.** *An allocation (c, y) pareto dominates another allocation $(\hat{c}, \hat{y})$ if*

- $c_i \succeq_i \hat{c}_i$ *for all* i
- $c_j \succ_j \hat{c}_j$ *for at least one* j

Definition 28 (Pareto Efficient)**.** *An allocation is pareto efficient if it is feasible and there is no alternative feasible allocation that pareto dominates it.*

There two important problems that allow us to think about reaching efficient allocations from the perspective of a social planner. Both may or may not lead to the same outcomes. I will let the exercises below speak more to exporing the depth behind each of these problems and just define it here for now.

Definition 29 (Arrow / Planner's Problem (for pure endowment economies))**.** *The arrow / planner's problem for agent i, given $\bar{u}_j$ for all $j \in \mathcal{M}$ such that $j \neq i$, is the following maximization problem.*

$$\begin{aligned} \max_{c_i \in C_i} \quad & U_i(c_i) \\ s.t. \quad (1) \quad & \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m}, \\ (2) \quad & U_j(c_j) \geq \bar{u}_j, \ \forall j \neq i. \end{aligned}$$

Definition 30 (Arrow / Planner's Problem (for production economies))**.** *The arrow / planner's problem for agent i, given $\bar{u}_j$ for all $j \in \mathcal{M}$ such that $j \neq i$, is the following maximization problem.*

$$\begin{aligned} \max_{c_i \in C_i} \quad & U_i(c_i) \\ s.t. \quad (1) \quad & \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m} + \sum_{h \in \mathcal{H}} y_{h,m}, \ \forall m \in \mathcal{M}, \\ (2) \quad & U_j(c_j) \geq \bar{u}_j, \ \forall j \neq i. \end{aligned}$$

Definition 31 (Negishi / Social Planner's Problem (for pure endowment economies))**.** *The negishi / social planner's problem, given weights λ_i for all $i \in \mathcal{I}$ is the following maximization problem*

$$\begin{aligned} \max_{c \in C} \quad & \sum_{i \in \mathcal{I}} \lambda_i U_i(c_i) \\ s.t. \quad & \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m}, \ \forall m \in \mathcal{M}. \end{aligned}$$

Definition 32 (Negishi / Social Planner's Problem (for production economies))**.** *The negishi / social planner's problem, given weights λ_i for all $i \in \mathcal{I}$ is the following maximization problem*

$$\begin{aligned} \max_{c \in C} \quad & \sum_{i \in \mathcal{I}} \lambda_i U_i(c_i) \\ s.t. \quad & \sum_{i \in \mathcal{I}} c_{i,m} \leq \sum_{i \in \mathcal{I}} e_{i,m} + \sum_{h \in \mathcal{H}} y_{h,m}, \ \forall m \in \mathcal{M}. \end{aligned}$$

3.3 Equilibrium Concepts

My biggest takeaway from this class was thinking more deeply about what an equilibrium or solving an economy means. I personally come from a macro background and very much just accepted the fact that a competive equilibrium is nothing more than a set of objects that satisfy another set of conditions. It just so happens to be the generally agreed upon solution concept but there are many more types of solution concepts that we could define.

For example, consider a simple pure endowment economy. One equilibrium concept that we could define is an autarky equilibrium where every agent consumes their endowment and prices are set such that no trade occurs. This too can be considered an equilibrium, we just tend to focus on competitive equilibrium. One other equilibrium concept that Phelan talks a lot about though is what he calls Jungle Equilibrium. It is not likely to show up in a prelim problem, but it is still an interesting concept to think about.

Definition 33 (Jungle Equilibrium)**.** *Fix some ordering of agents from weakest to strongest. Specifically, an ordering S where $i\,S\,j$ means agent i is stronger than agent j where if $i\,S\,j$ then not $j\,S\,i$. An allocation such that no agent can assemble a more preferred bundle by combining his own bundle either with a bundle that is held by one of the agents weaker than him or with the bundle that is freely available. Formally, it is a consumption allocation $c = \{\{c_{i,m}\}_{i\in\mathcal{I}}\}_{m\in\mathcal{M}}$ such that*

- $c_i \in C_i$*, for all* $i \in \mathcal{I}$.
- $\sum_i c_{i,m} \leq e_m$*, for all* $m \in \mathcal{M}$.
- *For all* $i, j \in \mathcal{I}$ *such that* $i\,S\,j$*. We have that* $c_i \succeq_i \hat{c}_i$ *for all* $\hat{c}_i \in C_i$ *such that* $\hat{c}_{i,m} \leq c_{i,m} + c_{j,m}$ *for all* $m \in M$.

Beyond this, the focus of the class was on exploring competitive equilibria and some basic important concepts (such as interpreting goods in different time periods as inherently different goods, or seeing how ownership plays with preferences (3.5.1)) that each of the exercises below will address.

Definition 34 (Competitive Equilibrium (for pure endowment economies))**.** *Consider a pure endowment economy. A competitive equilibrium is a collection of allocations and prices (c, p) such that*

- $p \in \mathbb{R}^M_+$.
- *(Feasibility)* $c_i \in C_i$ *for all* i.
- *(Feasibility - Market Clearing)* $\sum_{i\in\mathcal{I}} c_{i,m} \leq \sum_{i\in\mathcal{I}} e_m$ *for all* $m \in \mathcal{M}$.
- *(Feasibility - Budget Constraint)* $\sum_{m\in\mathcal{M}} p_m c_{i,m} \leq \sum_{m\in\mathcal{M}} p_m e_{i,m}$ *for all* $i \in \mathcal{I}$.
- *(Optimality) For any* $i \in \mathcal{I}$*, if* $\hat{c}_i \succ c_i$ *then* $\sum_{m\in\mathcal{M}} p_m \hat{c}_{i,m} > \sum_{m\in\mathcal{M}} p_m e_{i,m}$

Definition 35 (Competitive Equilibrium (for production economies))**.** *Consider a production economy. A competitive equilibrium is a collection of consumption allocations, firm production plans, and prices (c, y, p) such that*

- $p \in \mathbb{R}^M_+$.
- *(Feasibility)* $c_i \in C_i$ *for all* i.
- *(Feasibility)* $y_h \in Y_h$ *for all* h.
- *(Feasibility - Market Clearing)* $\sum_{i\in\mathcal{I}} c_{i,m} \leq \sum_{h\in\mathcal{H}} y_h + \sum_{i\in\mathcal{I}} e_m$ *for all* $m \in \mathcal{M}$.
- *(Feasibility - Budget Constraint)* $\sum_{m\in\mathcal{M}} p_m c_{i,m} \leq \sum_{m\in\mathcal{M}} p_m e_{i,m} + \sum_{h\in\mathcal{H}} \theta_{i,h} \sum_{m\in\mathcal{M}} p_m y_{h,m}$ *for all* $i \in \mathcal{I}$.

- *(Optimality - Household) For any $i \in \mathcal{I}$, if $\hat{c}_i \succ c_i$ then $\sum_{m\in\mathcal{M}} p_m \hat{c}_{i,m} > \sum_{m\in\mathcal{M}} p_m e_{i,m} + \sum_{h\in\mathcal{H}} \theta_{i,h} \sum_{m\in\mathcal{M}} p_m y_{h,m}$.*

- *(Optimality - Firm) If $\hat{y}_h$ is such that $\sum_{m\in\mathcal{M}} p_m \hat{y}_{h,m} < \sum_{m\in\mathcal{M}} p_m y_{h,m}$, then $\hat{y}_h \notin Y_h$.*

As a final note, a decent chunk of Phelan's class was dedicated to proving the welfare theorems for the two different types of economies listed above. I'm just going to list them for now since he doesn't usually ask for proofs.

Theorem 3.1 (First Welfare Theorem (for pure endowment economies))**.** *Let preferences be locally non-satiatable for every agent. If (p, c) is a competitive equilibrium, then c is pareto efficient.*

Theorem 3.2 (Second Welfare Theorem (for pure endowment economies))**.** *Let preferences be continous, convex, and strongly monotone for every agent. If c is pareto efficient (where $c >> 0$), then there exists some e, p such that (p, c) is a competitive equilibrium.*

Theorem 3.3 (First Welfare Theorem (for production economies))**.** *Let preferences be locally non-satiatable for every agent. If (p, c, y) is a competitive equilibrium, then (c, y) is pareto efficient.*

Theorem 3.4 (Second Welfare Theorem (for production economies))**.** *Let preferences be continous, convex, and strongly monotone for every agent. If (c, y) is pareto efficient (where $c >> 0$), then there exists some e, θ, p such that (p, c, y) is a competitive equilibrium.*

3.4 Introducing Uncertainty

As I have pointed out in this section so far. We can interpret uncertainty (and time periods for that matter) as similar to adding different types of goods. It is actually not all that different. For example, a banana in a world in which a coin has flipped heads is different than a banana in a world in which a coin has flipped tails are different goods. The math doesn't care that these are qualitatively different states as long as we know that they are treated differently. I'm not going to formalize everything and just hope that it shows up in some of the exercises below.

3.5 Exercises

3.5.1 Prelim: Spring 2025

Consider an economy with 2 goods, grapes and wine, and two agents, Amy and Barney. Initially, there exists 1 grape. There exist two firms which can turn grapes into wine according to the production function $w = 2\sqrt{g}$, where g is the quantity of grapes put into the production function, and w is the amount of wine produced. Amy only likes grapes. Barney only likes wine. Each prefer more to less and each has an ability to consume any non-negative quantity of either good.

(a) What are the production and consumption sets?

SOLUTION: The production set is

$$P = \{(-g, w) \in \mathbb{R}^2 : 0 \le g \le 1, \, 0 \le w \le 2\sqrt{g}\}.$$

The consumption set, for each agent $i \in \{Amy, Barney\}$,

$$C_i = \{(g, w) \in \mathbb{R}^2_+\}.$$

■

(b) Define an allocation given the environment described above. When is such an allocation feasible?

SOLUTION: Let $(g_A, w_A) = Z_A \in C_{Amy}$ and $(g_B, w_B) = Z_B \in C_{Barney}$ represent a particular consumption bundle for Amy and Barney respectively, and $(-g_1, w_1) = H_1 \in P, (-g_2, w_2) = H_2 \in P$ be elements of the production set for firm 1 and firm 2 respectively (given that they are both identical firms).

An **allocation** is the collection $\{Z_A, Z_B, H_1, H_2\}$. An allocation is **feasible** in this economy when, $g_A + g_B + g_1 + g_2 \le 1$ and $w_A + w_B \le w_1 + w_2$. ■

(c) Characterize the set of Pareto efficient allocations.

SOLUTION: Since each consumer prefers only different goods and grapes necessarily must be used to produce wine we can classify the set of pareto efficient allocations as

$$\boxed{\{(Z_A, Z_B, H_1, H_2) \,:\, g_B = 0, \, w_A = 0, \, g_A + g_1 + g_2 = 1, \, w_B = 2(\sqrt{g_1} + \sqrt{g_2})\}\,.}$$

■

(d) For general specifications of ownership of the initial grapes and the firms, define a competitive equilibrium. Will the allocation associated with any competitive equilibrium be efficient?

SOLUTION: A **competitive equilibrium** in this economy is a set of the collection of allocations $\{Z_A, Z_B, H_1, H_2\}$ and prices $\{p_g, p_w\}$ such that, given initial endowments of grapes e_A, e_B (where $e_A + e_B = 1$) and ownership of firm profits $\theta_{A1}, \theta_{A2}, \theta_{B1}, \theta_{B2}$ (where $\theta_{A1} + \theta_{B1} = \theta_{A2} + \theta_{B2} = 1$) such that

- $Z_A, Z_B \ge 0$.
- $p_g, p_w \in \mathbb{R}_+$

- (Grapes Market Clearing) $Z_A(g) + Z_B(g) \leq e_A + e_B + H_1(g) + H_2(g)$
- (Wine Market Clearing) $Z_A(w) + Z_B(w) \leq H_1(w) + H_2(w)$
- (Budget Constraint) $p_g Z_i(g) + p_w Z_i(w) \leq p_g e_i + \theta_{i1}(p_g H_1(g) + p_w H_1(w)) + \theta_{i2}(p_g H_2(g) + p_w H_2(w))$ for $i \in \{A, B\}$
- For all other agent allocations $\{\hat{Z}_A, \hat{Z}_B\}$ that are prefered to $\{Z_A, Z_B\}$, it must be that $\{\hat{Z}_A, \hat{Z}_B\}$ violates the budget constraint
- For all other firm allocations $\{\hat{H}_1, \hat{H}_2\}$, we have that it makes less profit for the firms.

Since this is a complete market, we can use the first welfare theorem and indeed conclude that any allocation associated with competitive equilibrium will be efficient.

■

(e) Find the set of competitive equilibria if Barney initially owns the grapes and Amy owns the firms?

Solution: Let's first look at the firm's problem for each firm $i \in \{1, 2\}$.

$$\max_{g_i} \quad p_w(2\sqrt{g_i}) - p_g g_i$$

Taking the first order condition, we get that

$$\frac{p_w}{\sqrt{g_i}} - p_g = 0$$
$$\implies g_i = \left(\frac{p_w}{p_g}\right)^2$$

This then also yields profits $\pi_i = \frac{p_w^2}{p_g}$. With this, we can now consider both of Amy and Barney's maximization problems. For Barney, we know that he is initially endowed with all of the grapes and he only likes wine. This means that they make income p_g and uses all of that to buy wine at price p_w. Barney's consumption of wine is p_g/p_w (and he consumes 0 grapes).

For Amy, we know that her only source of income is through the firm of the profits and she like's to consume grapes at unit cost p_g. As such, we know that her consumption of grapes is $2\frac{p_w^2}{p_g^2}$ (and she consumes 0 units of wine).

Finally, to pin down the prices, we use the market clearing condition for grapes. We know that the sum of the grapes that Amy consumes and the grapes the firms use to produce wine must equal 1.

$$\begin{aligned}
2\frac{p_w^2}{p_g^2} + 2\frac{p_w^2}{p_g^2} &= 1, \\
\frac{p_w^2}{p_g^2} &= \frac{1}{4}, \\
\frac{p_w}{p_g} &= \frac{1}{2}, \\
2p_w &= p_g.
\end{aligned}$$

Setting $p_w = 1$ as the numeraire, we get that $p_g = 2$. Putting everything together, we can finally classify the competitive equilibrium as

$$\boxed{Z_A = (0, 2),\ Z_B = (1/2, 0),\ H_1 = H_2 = (-1/4, 1),\ p_g = 2,\ p_w = 1.}$$

We can get the whole set of equilibria by shifting the numeraire. ■

(f) Find the set of competitive equilibria if Amy initially owns the grapes and Barney owns the firms?

SOLUTION: A simple trap that people may fall into is conclude that Amy consumes all of her endowment of grapes simply because she prefers only grapes and has no reason to sell her endowment. This is not how competitive equilibrium works! The agents in the economy don't know that no one else has no endowment of grapes or any of their characteristics. Agents only interact with prices and the wealth generation from the value that prices imbues on their endowment. So even if this may end up being the case, the reason is not because Amy knows specific information about other people and refuses to share her endowment.

In this case, we actually get, however, that **there is no competitive equilibrium**.

To show this, note first that the firm's problem is the exact same as before. Now, we know that Barney get's the profits and consumes only wine, this means that he consumes $2\frac{p_w}{p_g}$ units of wine. For Amy, she sells her entire endowment of grapes at price p_g and uses it to buy only grapes, which means that she consumes $p_g/p_g = 1$ units of grapes.

To pin down prices, consider the market clearing condition for grapes

$$1 + 2\frac{p_w^2}{p_g^2} = 1,$$

$$\frac{p_w^2}{p_g^2} = 0.$$

Setting a numeraire of $p_g = 1$ we get that the only possible value for the price of wine is $p_w = 0$. This, however, is not possible because a price of 0 for wine would entail that Barney would buy an infinite amount of wine while firms are producing no wine (given that Amy consumes all of the grapes). This violates the market clearing for wine. As such, there is no possible competitive equilibrium in this market. ■

4 ECON 8103 (Rustichini)

4.1 Exercises

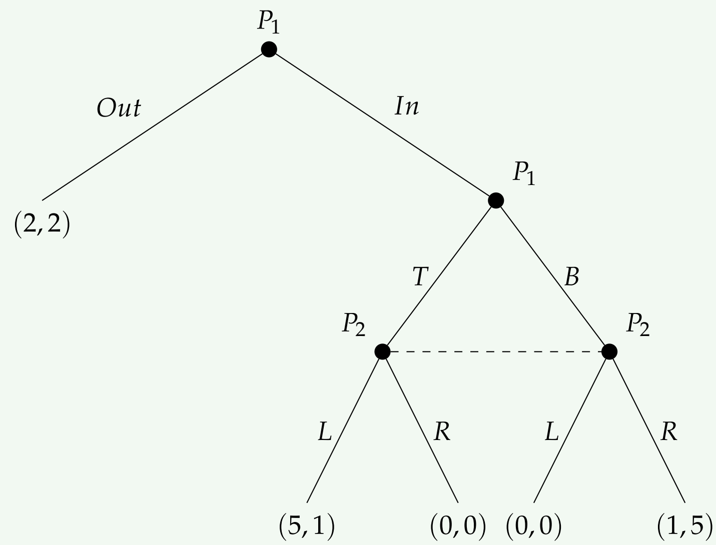
4.1.1 Prelim: Fall 2024

Consider the following game with two players. Player 1 moves first and chooses an action in the set $\{Out, In\}$. If he chooses Out the game is over and the two players receive a utility of $(2, 2)$, with the first component paid to the first player. If Player 1 chooses In, then the two players play a matrix game called G, where each player has a two action set, $\{T, B\}$ for the first player and $\{L, R\}$ for the second. Utilities in G are:

	L	R
T	$5, 1$	$0, 0$
B	$0, 0$	$1, 5$

(1) Write the game in extensive form.

Solution: This is the game in extensive form:



∎

(2) Write the game in normal form.

SOLUTION: This is the game in normal form. Player 1's strategies are $\{OT, OB, IT, IB\}$ (e.g. OT means "play Out, and play T if In is reached") and Player 2's strategies are $\{L, R\}$:

	L	R
OT	$(2,2)$	$(2,2)$
OB	$(2,2)$	$(2,2)$
IT	$(5,1)$	$(0,0)$
IB	$(0,0)$	$(1,5)$

∎

(3) Find all the Nash equilibria of the game.

SOLUTION: Looking at the normal form game, the Nash equilibria are

$$\boxed{\{(OT, R),\ (OB, R),\ (IT, L)\}.}$$

	L	R
OT	$2,\mathbf{2}$	$\mathbf{2},\mathbf{2}$
OB	$2,\mathbf{2}$	$\mathbf{2},\mathbf{2}$
IT	$\mathbf{5},\mathbf{1}$	$0,0$
IB	$0,0$	$1,\mathbf{5}$

■

(4) Find all the subgame perfect equilibria of the game.

Solution: There are two subgames: the entire game itself, and the subgame that occurs if Player 1 chooses In. We already know the Nash Equilibria of the entire game, so let us consider the In subgame:

	L	R
T	$5,1$	$0,0$
B	$0,0$	$1,5$

The Nash equilibria of the subgame are $\{(T,L),\ (B,R)\}$. Matching these with the Nash Equilibria of the full game above, the subgame perfect equilibria of the game are

$$\boxed{\{(IT,L),\ (OB,R)\}.}$$

■

(5) Find the game resulting from iterated elimination of strictly dominated strategies.

Solution: Writing down the normal form game and crossing out strategies that are strictly dominated:

	L	R
OT	$2,2$	$2,2$
OB	$2,2$	$2,2$
IT	$5,1$	$0,0$
$\cancel{IB}$	$\cancel{0,0}$	$\cancel{1,5}$

IB is strictly dominated by OT (Player 1 always receives 2 from OT versus at most 1 from IB). No further strict dominations exist, so the resulting game is:

	L	R
OT	2,2	2,2
OB	2,2	2,2
IT	5,1	0,0

■

(6) Find the games resulting from iterated elimination of weakly dominated strategies.

Solution: We proceed in the following steps:

(1) IT strictly dominates IB.

(2) L weakly dominates R (for Player 2).

(3) IT strictly dominates OT and OB.

	L	$\cancel{R}$
$\cancel{OT}$	$\cancel{2,2}$	$\cancel{2,2}$
$\cancel{OB}$	$\cancel{2,2}$	$\cancel{2,2}$
IT	5,1	$\cancel{0,0}$
$\cancel{IB}$	$\cancel{0,0}$	$\cancel{1,5}$

The resulting game is:

	L
IT	5,1

■

(7) Suppose that in a pre-play communication players have agreed to play the following strategy profile: "Player 1 plays Out. Player 2 will play R in the event that the game G is reached."

Suppose that in the actual game player 2 is actually called to move. What should player 2 think that player 1 just did, and what should player 2 choose? What does the iterated elimination of weakly dominated strategies suggest?

Solution: Player 2 knows that by sticking to the agreed strategy, Player 1 would receive 2 units of utility. This means that Player 2 knows Player 1 would only deviate if they are intending to obtain more than 2 units of utility. As such, Player 2 assumes that Player 1 will play IT, and in order to obtain more than 0 units of utility, Player 2 will play L.

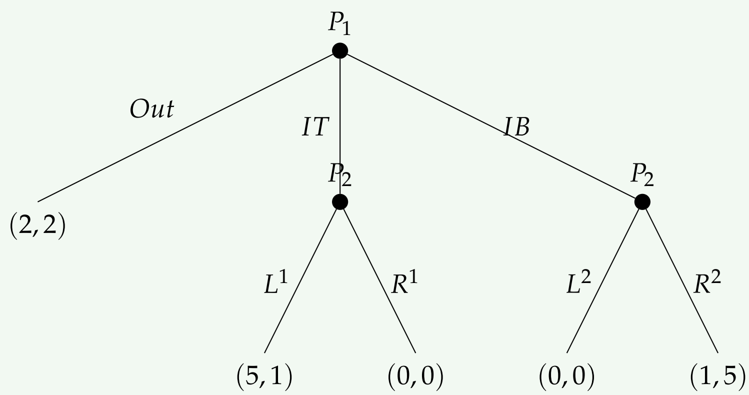
Iterated elimination of weakly dominated strategies also suggests the outcome

$$\boxed{(IT, L).}$$

■

(8) Consider the game with two players in which player 1 moves first and chooses an action in the set $\{Out, IT, IB\}$. If this player chooses Out then the game ends with payoffs $(2,2)$; if he chooses an action in $\{IT, IB\}$ then player 2 can choose an action in the set $\{L, R\}$ and payoffs are as in the game G with IT and IB replacing T and B respectively. Answer points 1–7.

Solution: **(1)** Here is the extensive form game:



(2) Here is the normal form game. Player 2 now has four strategies $\{L^1L^2,\ R^1L^2,\ L^1R^2,\ R^1R^2\}$, where the superscript denotes the action in the IT branch (^{1}) and the IB branch (^{2}):

	L^1L^2	R^1L^2	L^1R^2	R^1R^2
OUT	$(2,2)$	$(2,2)$	$(2,2)$	$(2,2)$
IT	$(5,1)$	$(0,0)$	$(5,1)$	$(0,0)$
IB	$(0,0)$	$(0,0)$	$(1,5)$	$(1,5)$

(3) Starting from the normal form, the Nash equilibria are:

$$\{(IT,\ L^1L^2),\ (OUT,\ R^1L^2),\ (IT,\ L^1R^2),\ (OUT,\ R^1R^2)\}.$$

	L^1L^2	R^1L^2	L^1R^2	R^1R^2
OUT	$2, \mathbf{2}$	$\mathbf{2}, \mathbf{2}$	$2, \mathbf{2}$	$\mathbf{2}, \mathbf{2}$
IT	$\mathbf{5}, \mathbf{1}$	$0, 0$	$\mathbf{5}, \mathbf{1}$	$0, 0$
IB	$0, 0$	$0, 0$	$1, \mathbf{5}$	$1, \mathbf{5}$

(4) There are three subgames: the entire game, the IT subgame, and the IB subgame. In the IT subgame, L^1 is an equilibrium (Player 2 gets $1 > 0$). In the IB subgame, R^2 is an equilibrium (Player 2 gets $5 > 0$). Matching these with the Nash equilibria of the full game above, the SPE is:

$$\boxed{(IT,\ L^1,\ R^2).}$$

(5) We use iterated elimination of strictly dominated strategies. OUT strictly dominates IB (Player 1 always receives 2 from OUT versus at most 1 from IB). No further strict dominations remain.

	L^1L^2	R^1L^2	L^1R^2	R^1R^2
OUT	$2, 2$	$2, 2$	$2, 2$	$2, 2$
IT	$5, 1$	$0, 0$	$5, 1$	$0, 0$
~~IB~~	~~$0, 0$~~	~~$0, 0$~~	~~$1, 5$~~	~~$1, 5$~~

(6) We use iterated elimination of weakly dominated strategies:

- OUT strictly dominates $IB \Rightarrow$ eliminate IB.
- L^1L^2 weakly dominates R^1L^2 (for Player 2) $\Rightarrow$ eliminate R^1L^2.
- L^1L^2 weakly dominates R^1R^2 (for Player 2) $\Rightarrow$ eliminate R^1R^2.
- IT strictly dominates $OUT \Rightarrow$ eliminate OUT.

	L^1L^2	~~R^1L^2~~	L^1R^2	~~R^1R^2~~
~~OUT~~	~~$2, 2$~~	~~$2, 2$~~	~~$2, 2$~~	~~$2, 2$~~
IT	$5, 1$	~~$0, 0$~~	$5, 1$	~~$0, 0$~~
~~IB~~	~~$0, 0$~~	~~$0, 0$~~	~~$1, 5$~~	~~$1, 5$~~

The resulting game after elimination is:

	L^1L^2	L^1R^2
IT	$5,1$	$5,1$

(7) Again, we know that if Player 1 is deviating from OUT, they are trying to obtain more than 2 units of utility and will be choosing IT (since IB gives at most 1). As such, Player 2 will then choose L^1, giving either (L^1, L^2) or (L^1, R^2). ■

5 ECON 8104 (Rahman)

5.1 Exercises